

Tasks :

Create mock software components (SWCs).

Exchange signals between components.

Log communication timing.

In a modern vehicle, infotainment features like volume control, GPS navigation, and dashboard displays are managed by independent Software Components (SWCs). These components do not talk to each other directly; instead, they communicate through a virtual layer called the Runtime Environment (RTE).

Key Concepts in this Simulator:

- **Virtual Functional Bus (VFB):** The simulator mimics the VFB, allowing components to exchange signals (like volume levels) without knowing the underlying hardware details.
 - **Layered Architecture:** It follows the AUTOSAR layered approach, separating the Application Layer (SWCs) from the Communication Layer (RTE).
 - **Deterministic Timing:** A critical requirement for automotive systems is **latency management**. This simulator measures and logs the exact time it takes for a signal to travel from the user interface (HMI) to the audio hardware, ensuring the system meets real-time performance requirements.
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1. Task Breakdown

Task A: Create Mock Software Components (SWCs)

In AUTOSAR, an SWC is an atomic unit of code. For an infotainment system, you might have:

- **MediaController:** Handles play/pause commands.
- **AudioOutput:** Manages physical sound levels.
- **HMI_Component:** Represents the touch screen user interface.

Task B: Exchange Signals (Ports & Interfaces)

Communication between SWCs is handled via **Ports** using **Sender-Receiver** or **Client-Server** patterns.

- **Sender:** **HMI_Component** sends a **Volume_Level** signal.
- **Receiver:** **AudioOutput** receives the signal and adjusts the driver.

Task C: Log Communication Timing

Since infotainment systems are performance-sensitive, you must track how long it takes for a signal to move from one component to another (End-to-End latency).

Flowchart:

